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## Motivation Letter for the MSc in Computer Science

Dear Members of the Selection Committee,

Since my first line of code in high school, where I was fed up doing calculations by hand for the olympiads in physics, till today, I have come to the conclusion that writing software is doable, but writing reliable and secure software is hard. As a result, I have devoted myself to researching ways to verify software without the need of human labor.

One method to automate this process is model checking in formal methods. That is the field I do my research on. Currently, I do this research at BME in the FTSRG research group, where we apply and test our software in the Theta framework using SV-COMP. I work on ways to pass the partial results from different model checking algorithms like CEGAR, K-induction, and IMC. These partial results happen when we decide on using some heuristics and AI models that a specific algorithm is not able to solve this verification task. I have had some good results on this, for which I also got first prize in the TDK (Student Scientific Conference) at BME. The reason I chose this field is that I love mathematics. Especially exact mathematics with solvers and graphs. I must mention, I just love a nice problem to think about. As a result, I also do a lot of competitive programming with the Computing Programming Group at BME. I also represented BME at ICPC CERC 2023.

When it comes to courses, I usually focus on security and networks, while I am also part of the database group at BME as a lab instructor. I have a lot of knowledge when it comes to internet protocols, especially in IPsec.

ETH Zurich has a lot of work done in this field, also especially in the Viper Verification Infrastructure with its frontend tools. Formal methods had a broader applicability than just verifying software, so there is also work in applying these formal methods in security, especially verifying important protocols with tools such as Tamarin Prover.

I dedicate my life to science and engineering. I spend nearly 12 hours every day in this field. People say I am gifted. I only gift I have is being alive and struggling towards something greater. I try to work the hardest to my humanly capable limits. People usually judge my route for being inhuman routine. But this constant struggle to reach the peak of research in an infinite large mountain, is what keeps me going in life. It gives me the experience of the beauty of truth and the beauty of creation.

Humans try to avoid change, because with change comes hardships to adapt. True, I could also continue my master's at BME Budapest where I am familiar with everything, and it would be quite easy. But I want to grow more, experience more, no matter how hard it is. And ETH Zurich is the place for my next journey, where it will allow me to do research in verifying software and also expand the formal methods field in the networks and security.

Sincerely,  
Klevis Imeri

November 30, 2025